

CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Schiebinger

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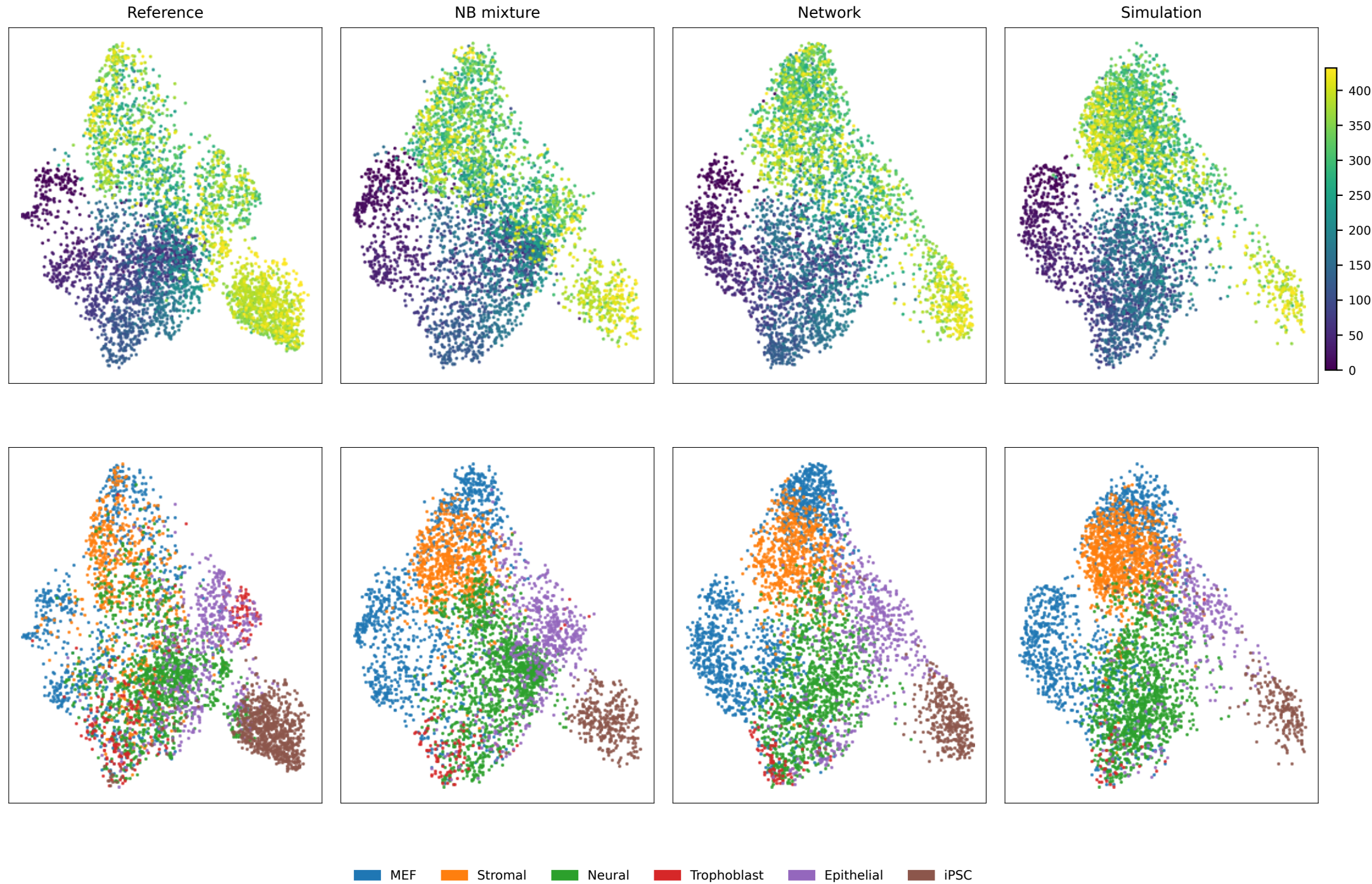
Run		Perturbations (Data/KO_OV_simulate.txt)	
Split	train	✓	OV Obox6
Stimulus penalisation	1.0	✓	OV Zfp42
Prior penalisation	1.0		
Genes	108		
Stimuli	2		
Observed cells	10804		
Timepoints	0, 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192,		
Cell types	204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432		
Network shown	inter_simul.npy, network #0 / 1		
UMAP preprocessing	normalise=False, log1p=True		

Contents

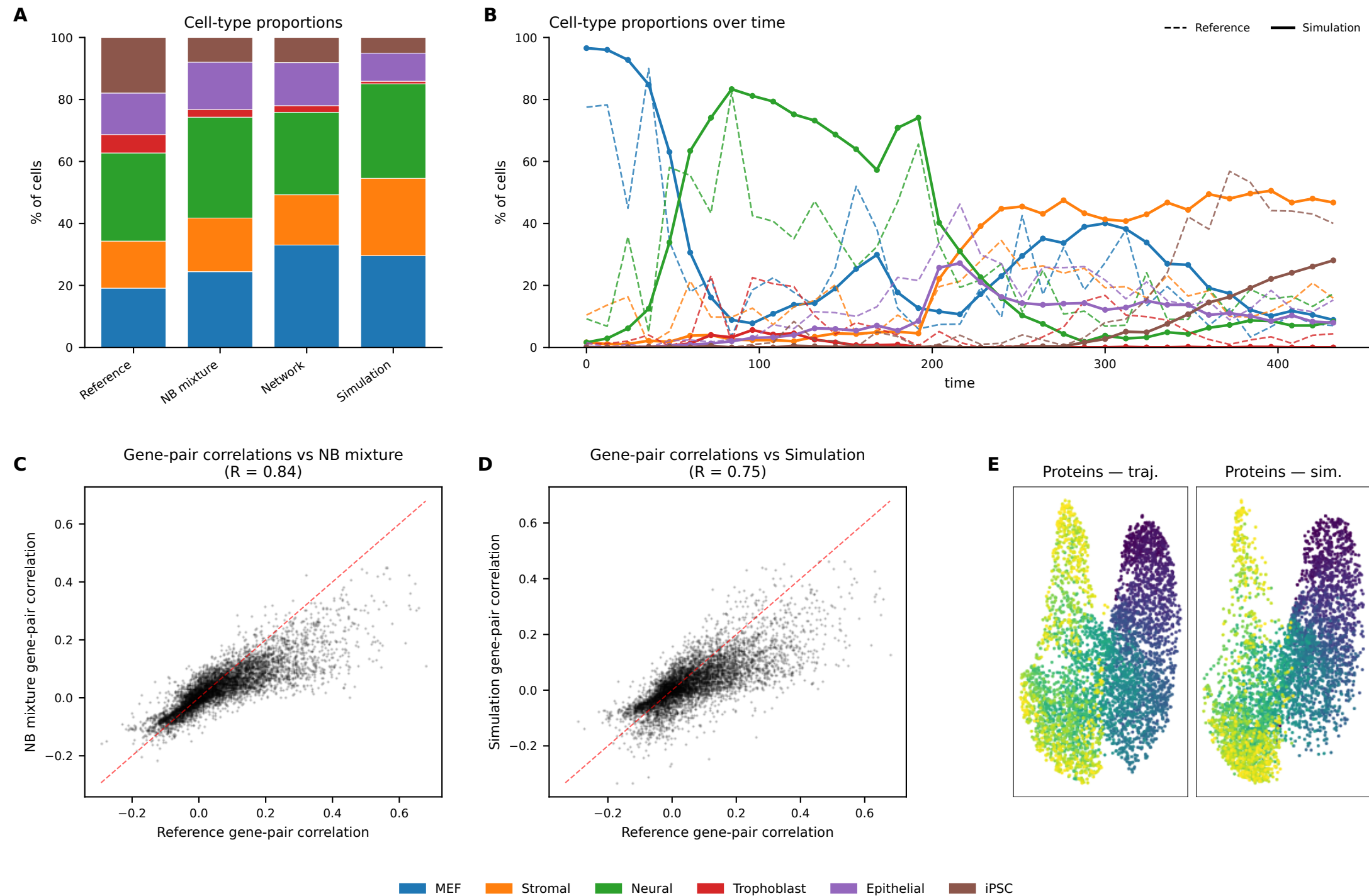
1. Generative model — UMAPs of data, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots) and top-10 regulators + stimulus
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth
5. Learned dynamics — mRNA and protein velocity fields (mechanistic and along trajectories)

1. Generative model — trajectories and simulation

Joint UMAP of the observed data (Reference), NB mixture, network-driven modes and full simulation.



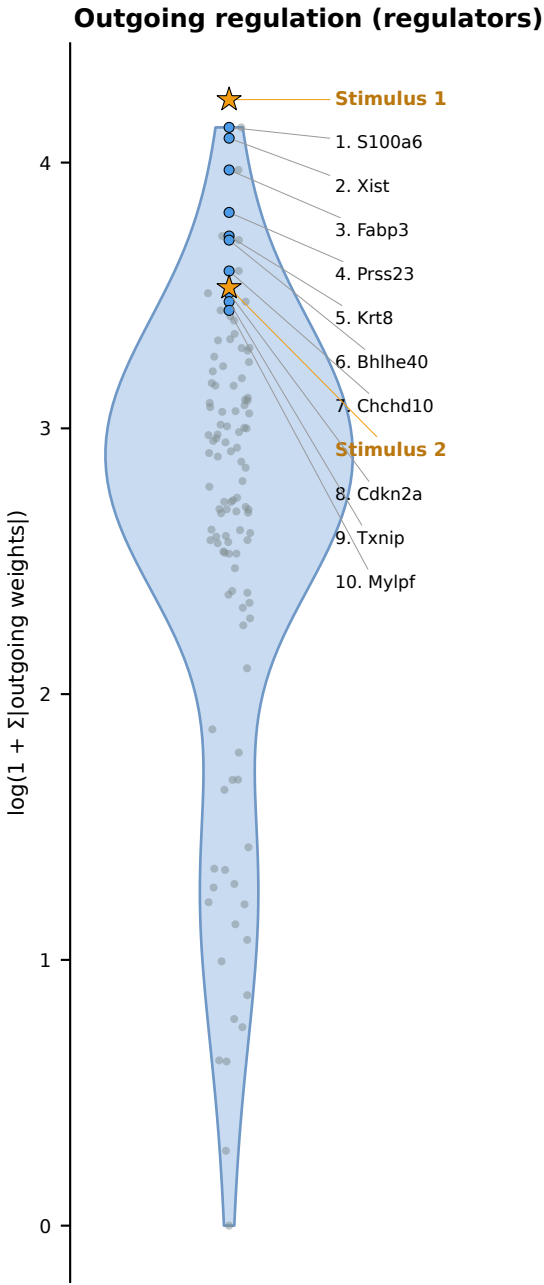
1. Generative model — quantitative checks



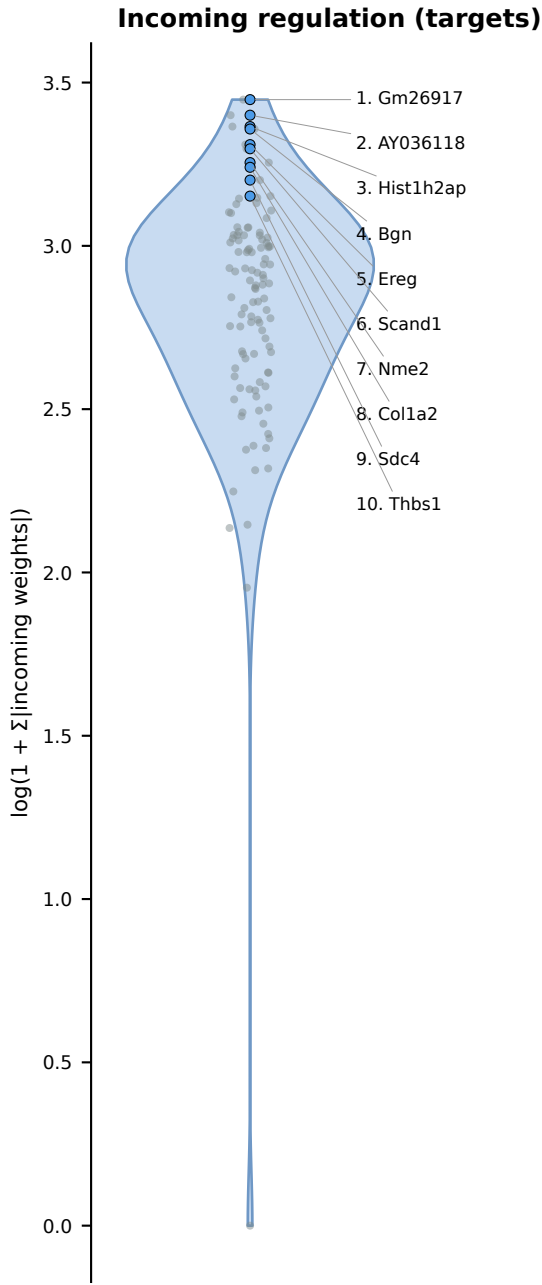
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 10 genes labelled; stimulus shown as a star (stimulus = 1.0 $\geq$ 0.5).

A



B



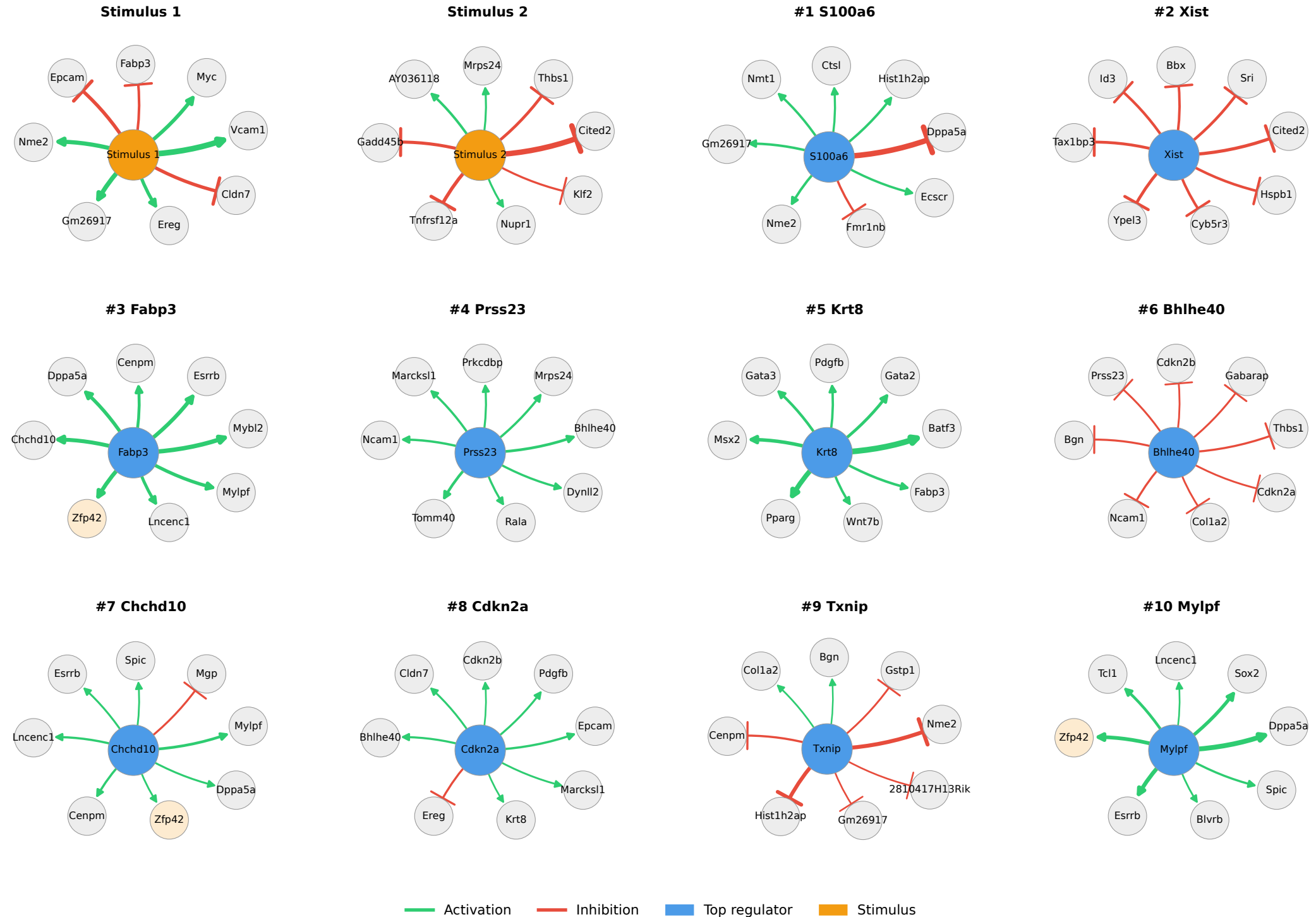
C

Top regulators

#	Regulator	$\Sigma w $	# act.	# inh.
1	S100a6	61.33	83	12
2	Xist	58.86	11	73
3	Fabp3	52.10	57	23
4	Prss23	44.26	74	18
5	Krt8	40.42	62	28
6	Bhlhe40	39.79	14	82
7	Chchd10	35.31	53	42
8	Cdkn2a	32.40	76	21
9	Txnip	31.36	54	43
10	Mylpf	30.31	46	23
★	Stimulus 1	68.20	43	48
★	Stimulus 2	33.11	48	18

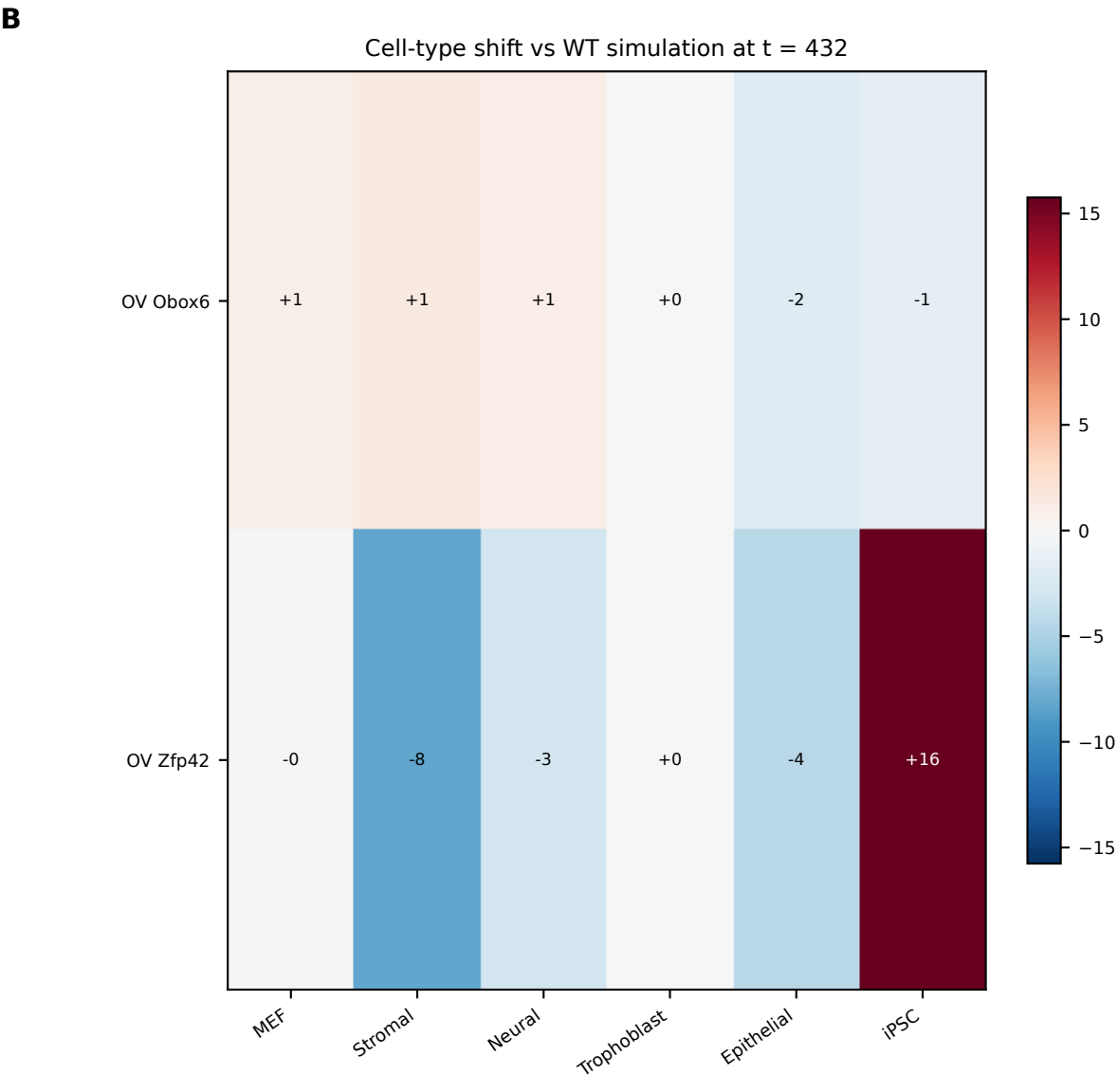
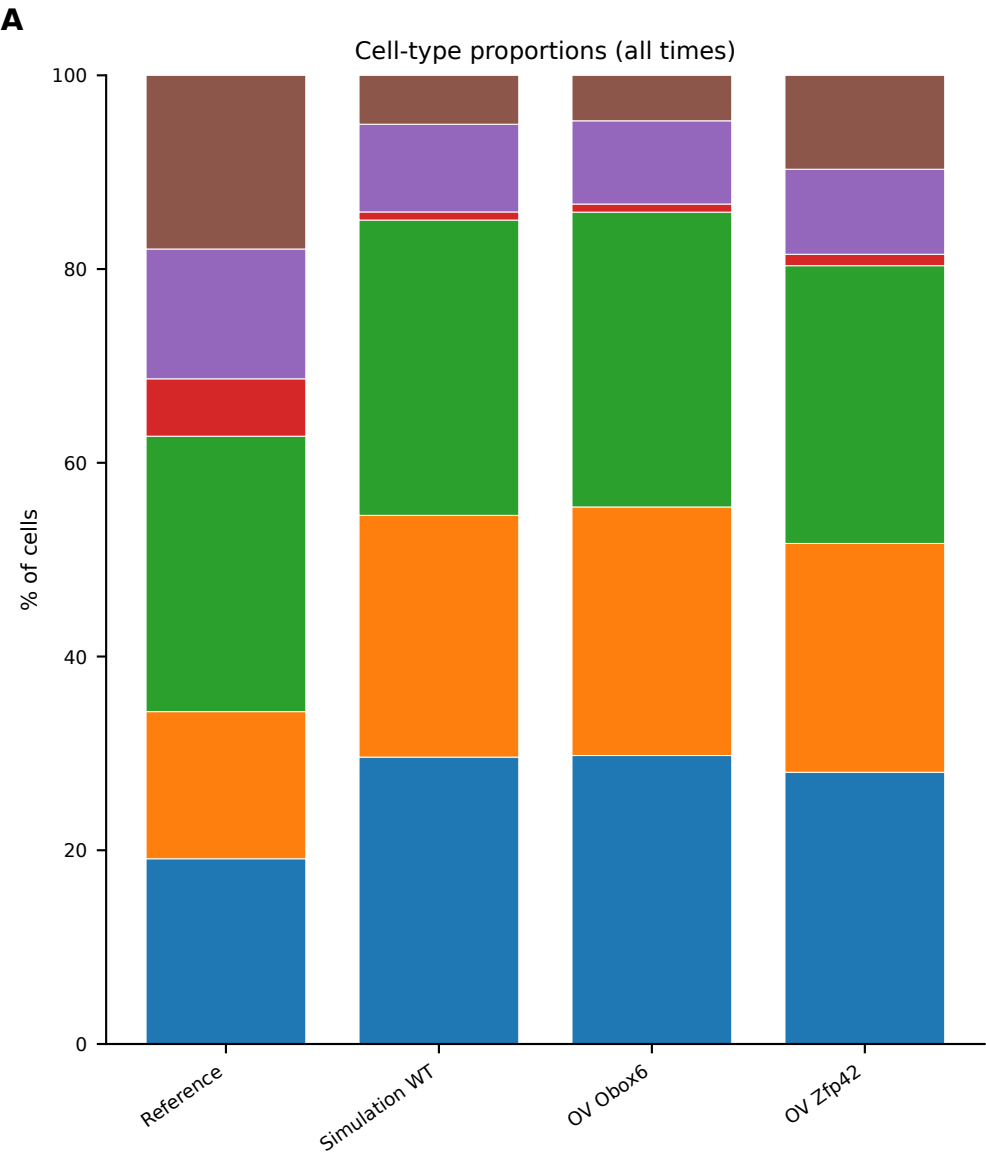
2. Gene regulatory network — top regulators and their main targets

Top 8 targets per regulator; edge width $\propto$ |weight| (global scale). Perturbed genes (KO/OV) are highlighted in beige.



3. In-silico perturbations — overview

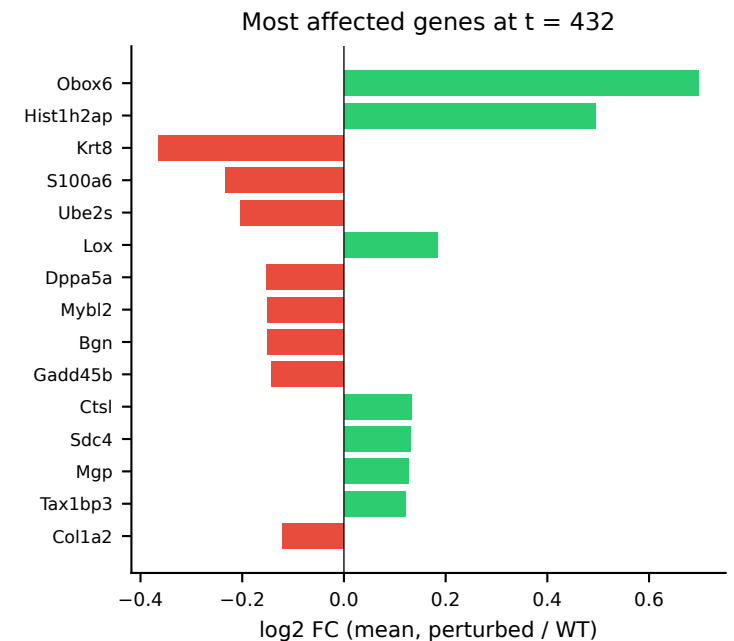
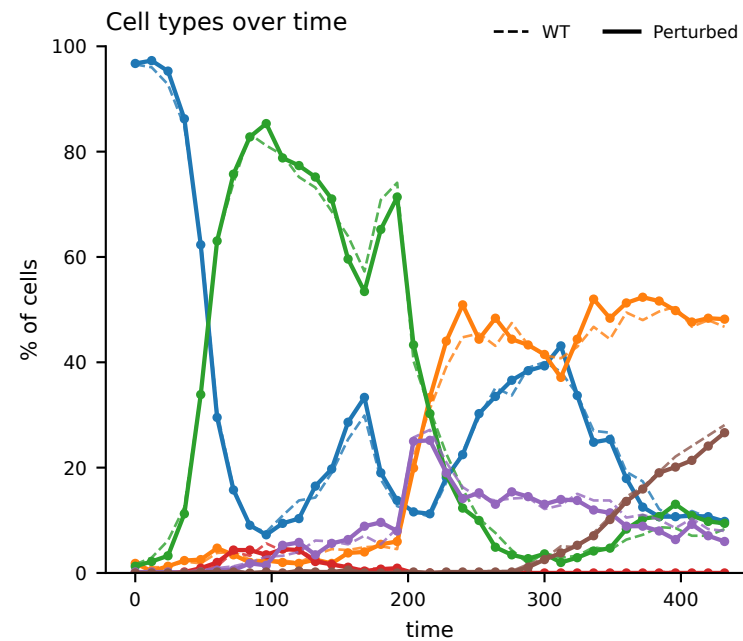
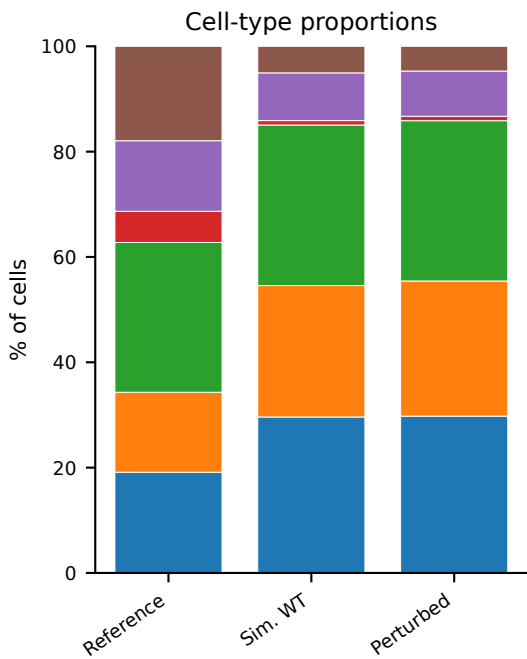
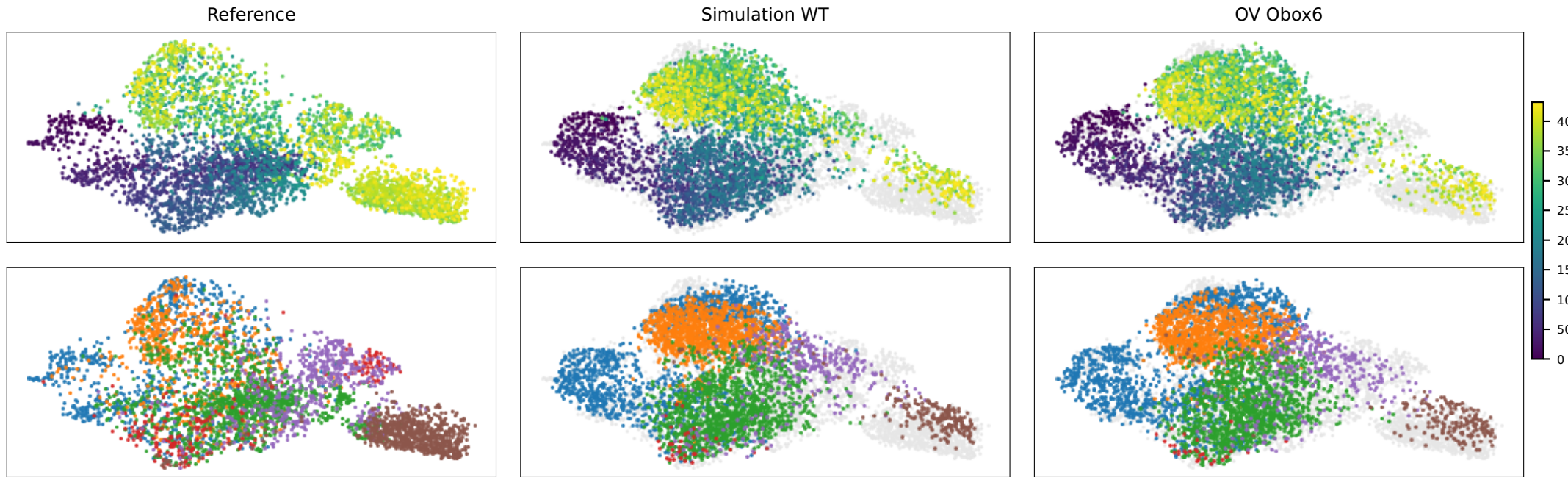
Cell types predicted by a random forest trained on the observed data.



MEF Stromal Neural Trophoblast Epithelial iPSC

3. Perturbation — OV Obox6

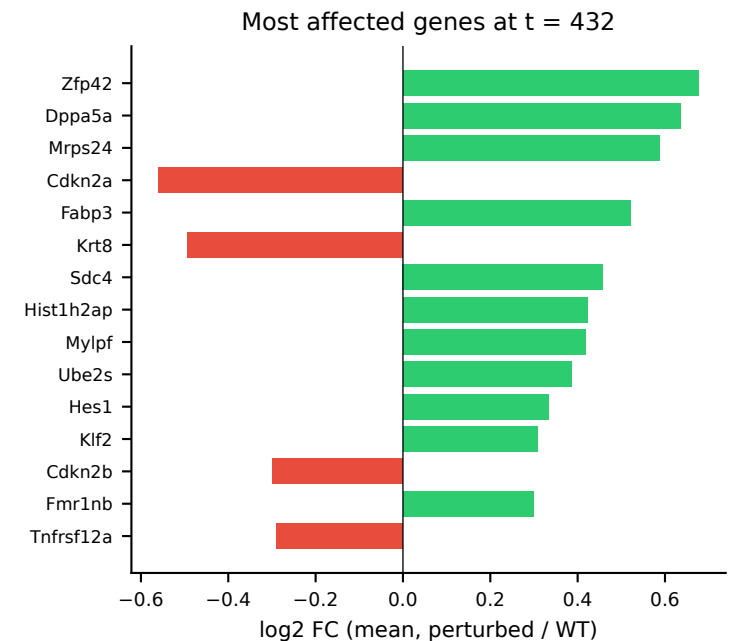
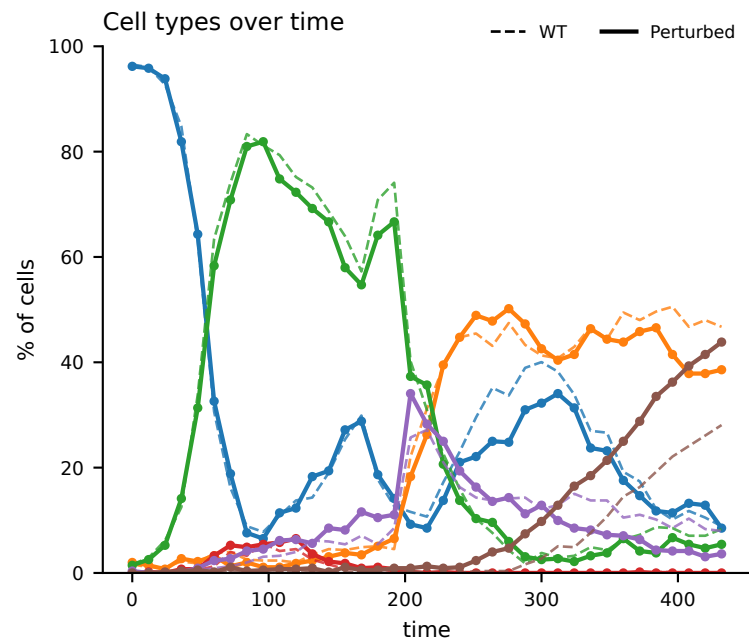
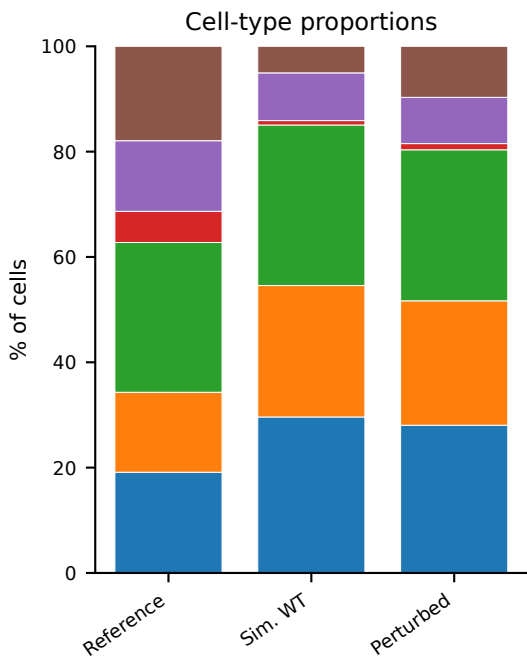
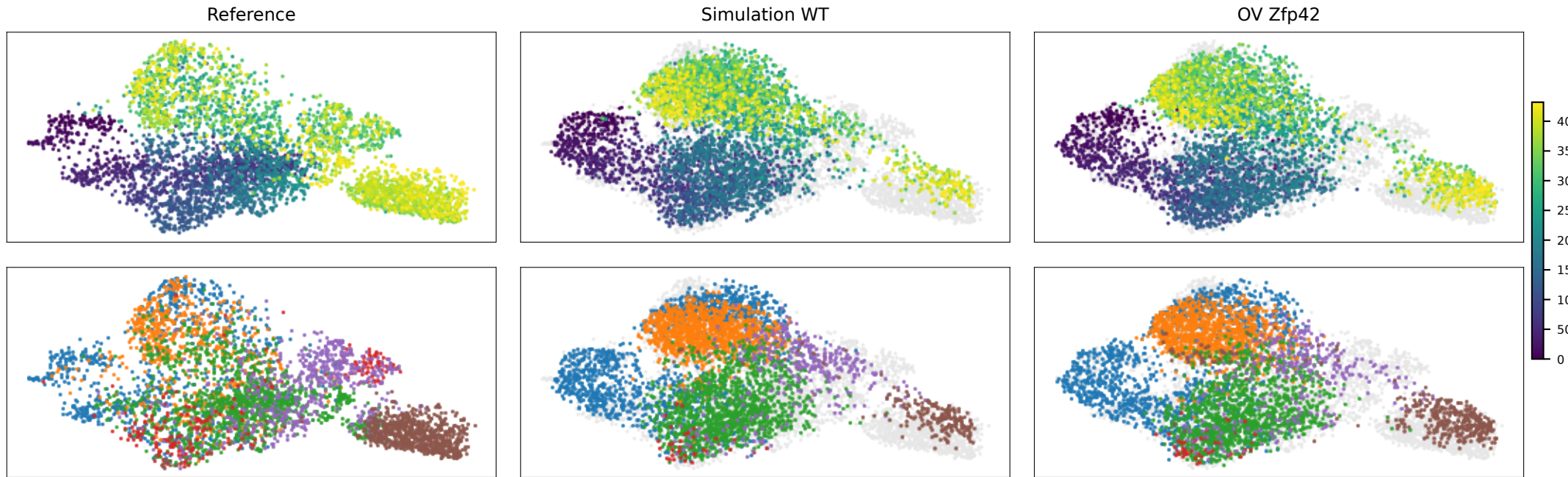
KO_none_OV_Obox6. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



MEF Stromal Neural Trophoblast Epithelial iPSC

3. Perturbation — OV Zfp42

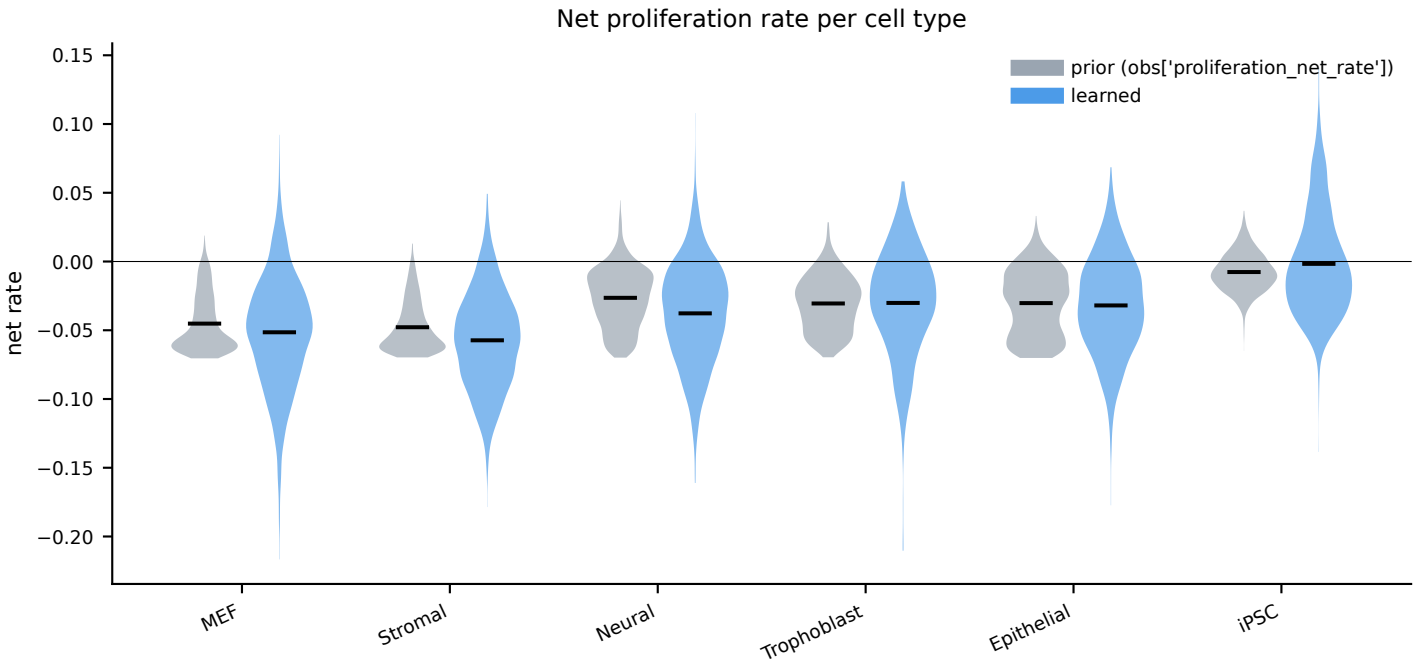
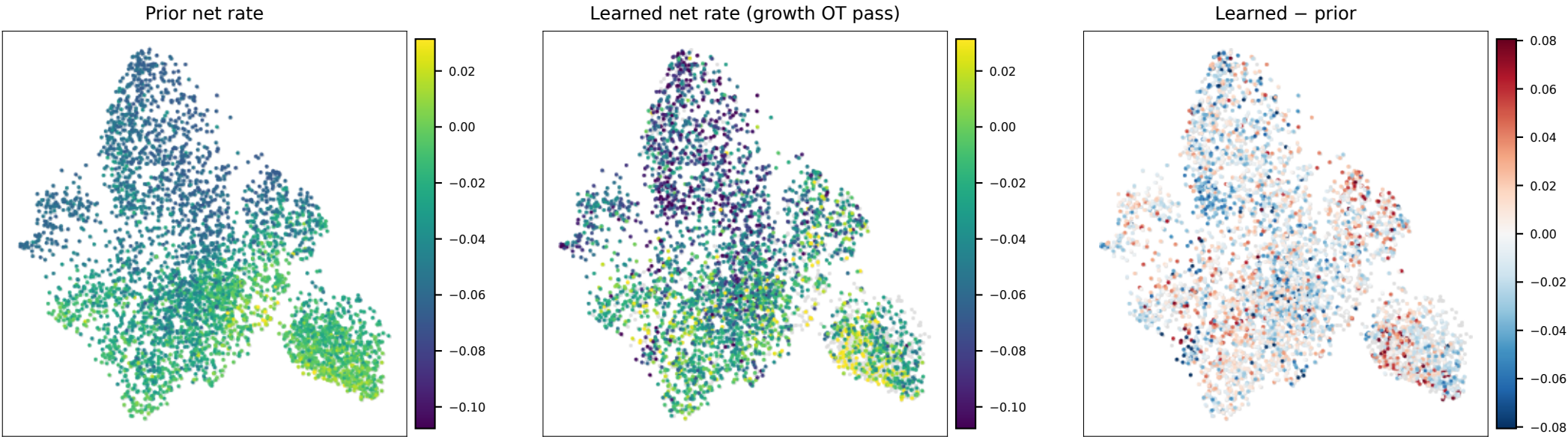
KO_none_OV_Zfp42. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



MEF Stromal Neural Trophoblast Epithelial iPSC

4. Proliferation — prior vs learned net rate (birth – death)

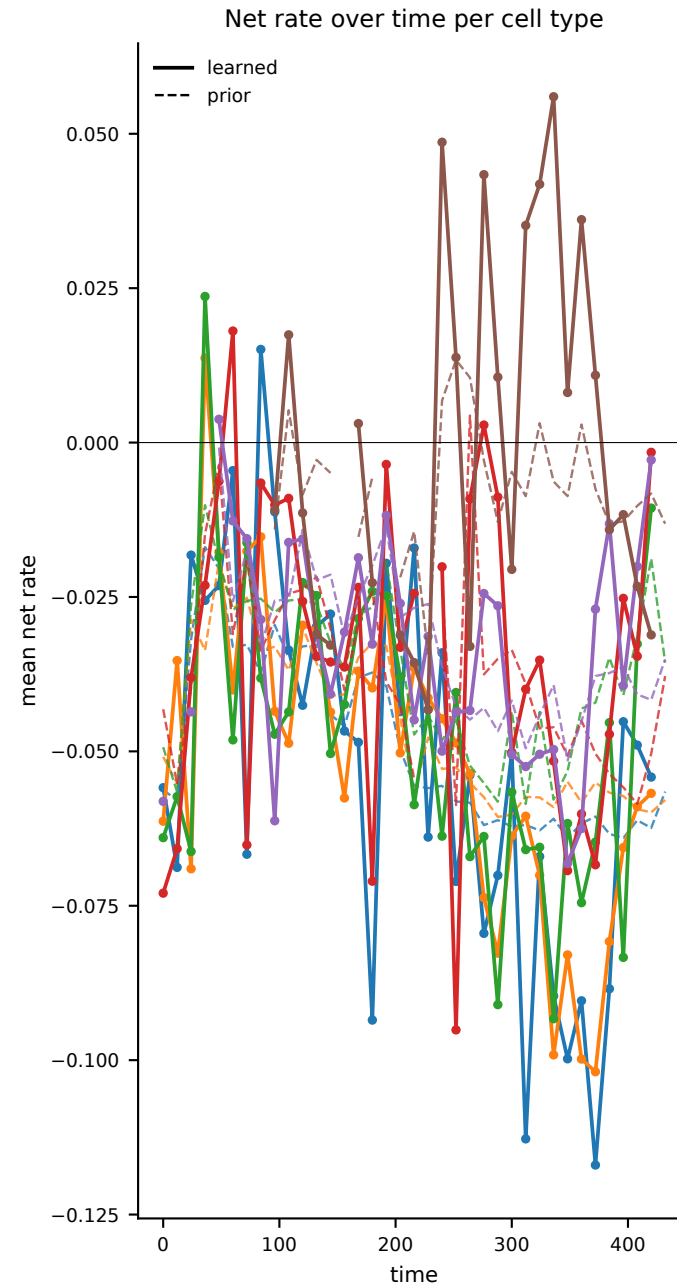
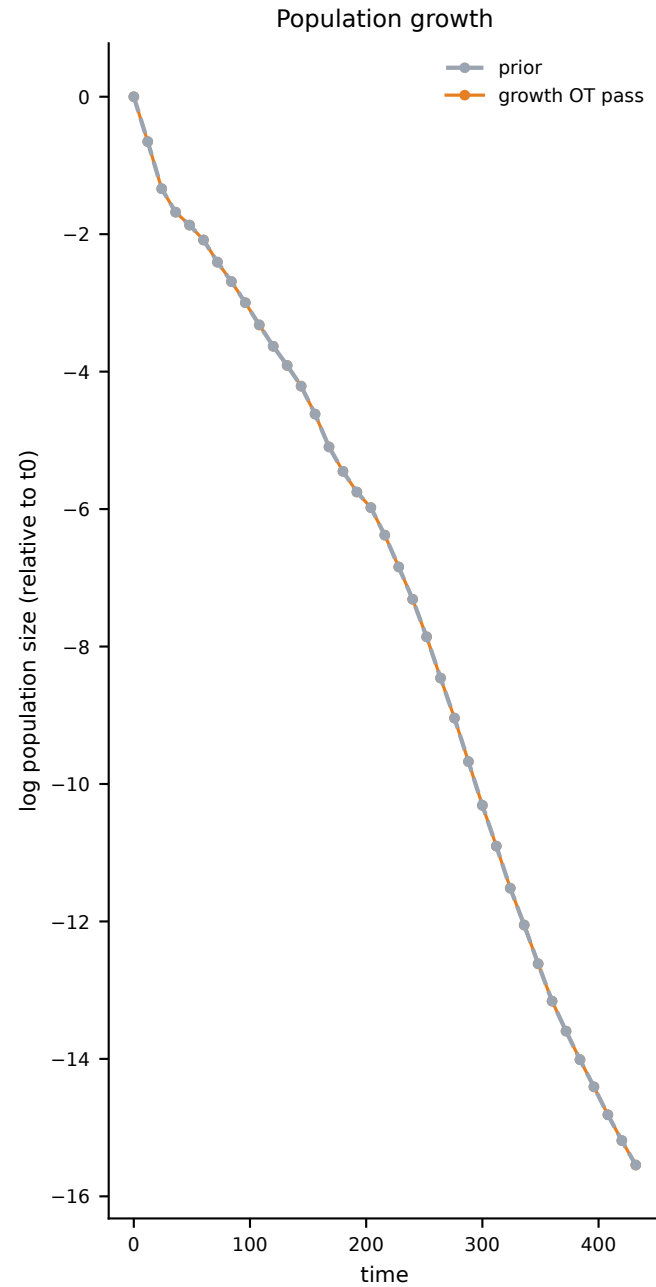
Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.



Mean net rate per cell type			
Cell type	Anchor	Prior	Learned
MEF	—	-0.0452	-0.0515
Stromal	—	-0.0478	-0.0573
Neural	—	-0.0264	-0.0377
Trophoblast	—	-0.0305	-0.0301
Epithelial	—	-0.0302	-0.0319
iPSC	—	-0.00766	-0.0017

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; unless Data/population_sizes is given, its absolute level follows the prior.

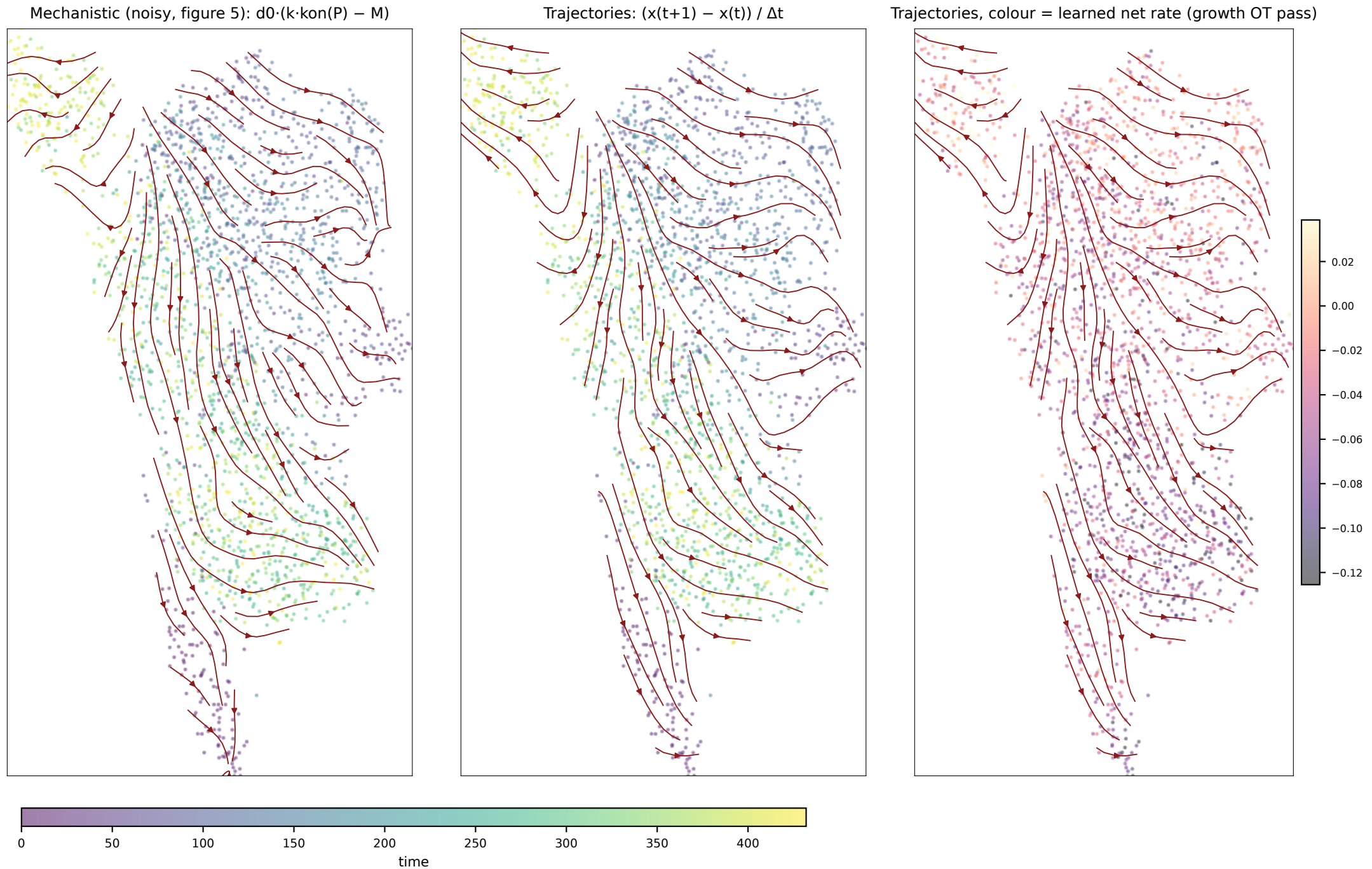


No proliferation MLP
(run infer_network_simul
--compute-proliferation)

MEF Stromal Neural Trophoblast Epithelial iPSC

5. Learned dynamics — mRNA velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of log1p NB-sampled mRNA. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.72.



5. Learned dynamics — Proteins velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of protein levels. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.51.

